

CHESS RULES

Implementation Reference for a Local 2-Player Chess Website

Purpose
This document is the chess-rules source of truth for implementing a real, two-player chess game on one shared device. It defines legal moves, special moves, check/checkmate, draws, and the game states required for correct rule enforcement.

1. Board and Initial Setup

Board: Chess is played on an 8×8 board containing 64 squares arranged in eight ranks and eight files.

Initial placement: From White's perspective, the back rank is arranged left-to-right as **Rook, Knight, Bishop, Queen, King, Bishop, Knight, Rook**. White's pawns occupy the second rank. Black uses the mirrored arrangement on the eighth rank, with Black's pawns on the seventh rank. The queens start on their own color.

- **White moves first.** Thereafter, turns alternate between White and Black.
- A turn consists of making one legal move for the side whose turn it is.
- A king may never remain in check after its player's move. A position in which a king is attacked is not a legal completed position for that side.

2. Piece Movement

King

- Moves exactly one square in any direction: horizontally, vertically, or diagonally.
- May capture an opposing piece on an adjacent square if the destination square is not attacked by an opposing piece.
- The king may not move onto an attacked square, move adjacent to the opposing king, or otherwise end the move in check.
- Castling is a special king move covered in Section 7.

Queen

- Moves any number of squares horizontally, vertically, or diagonally.
- The path must be clear; a queen cannot jump over any piece.
- Captures by moving to a square occupied by an opposing piece, provided the path is clear.

Rook

- Moves any number of squares horizontally or vertically.
- The path must be clear; a rook cannot jump over any piece.
- Captures an opposing piece on the destination square when the path is clear.
- Castling uses a rook as part of the special move described in Section 7.

Bishop

- Moves any number of squares diagonally.
- A bishop remains on the same square color for its entire lifetime.

- The path must be clear; a bishop cannot jump over any piece.

Knight

- Moves in an L-shape: two squares in one orthogonal direction and then one square perpendicular to that direction, for eight possible destinations.
- The knight may jump over other pieces.
- It captures an opposing piece by landing on its square.

Pawn

- Moves forward one square into an empty square. White pawns move toward Black's side; Black pawns move toward White's side.
- From its starting rank, a pawn may move forward two squares if both the intermediate square and destination square are empty.
- A pawn captures one square diagonally forward, onto a square occupied by an opposing piece, except for the special en passant capture.
- A pawn cannot capture by moving straight forward.
- A pawn that reaches the opposite end of the board must be promoted; see Section 9.

3. Legal and Illegal Moves

Legal move: A move is legal only if the moving piece follows its movement rules, the move does not violate occupancy/path restrictions, and after the move the moving player's king is not in check.

Illegal move: Any move that violates piece movement, attempts to capture a friendly piece, jumps when jumping is not permitted, uses a blocked path, moves the king into attack, or leaves the moving player's own king in check.

- A player may never make a move that leaves their own King in check.
- A player may not capture the opposing king. Checkmate ends the game before such a capture could occur.

State transition after a legal move: Apply the move to the board, remove any captured piece, apply special-move effects (castling, en passant, or promotion), update any relevant move-history information, and switch the side to move. Then evaluate the resulting position for check, checkmate, stalemate, and applicable draw conditions.

4. Check

Check means the king of the side to move is currently attacked by at least one opposing piece.

Detection: A king is in check if any opposing piece has a legal attack on its square, considering that piece's movement and blockers. For attack detection, pawn attack squares are the diagonally forward squares, knights attack their L-shaped destinations, kings attack adjacent squares, and sliding pieces attack along unobstructed lines.

A position in check is not automatically game over. The checked player must make a legal move that removes the check.

- Move the king to a square that is not attacked.
- Capture the checking piece, if the capture is legal and leaves the king safe.
- Block a sliding attack by placing a piece between the attacker and the king, when blocking is possible.
- For a double check, the only legal response is a king move.

5. Checkmate

Checkmate occurs when the side to move is in check and has **no legal move** that removes the check.

- The game ends immediately when checkmate is established.
- The player delivering checkmate wins; the checkmated player loses.
- The opposing king is never captured; the game ends at checkmate.

6. Stalemate

Stalemate occurs when the side to move has **no legal moves** and its king is **not in check**.

Stalemate ends the game and the result is a **draw**.

7. Castling

Castling is one legal move involving the king and a rook. The king moves two squares toward the rook, and that rook moves to the square immediately on the other side of the king.

Kingside castling

- White: king e1 → g1; rook h1 → f1.
- Black: king e8 → g8; rook h8 → f8.

Queenside castling

- White: king e1 → c1; rook a1 → d1.
- Black: king e8 → c8; rook a8 → d8.

Conditions required

- The king and the relevant rook must both be on their original squares and must not have previously moved.
- All squares between the king and rook must be empty.
- The king must not currently be in check.
- The king may not pass through a square that is attacked by an opposing piece.
- The king may not end on a square that is attacked by an opposing piece.
- The castling move must therefore not take the king out of, through, or into check.

Only the king's start square, transit square, and destination square are subject to the attack restriction. The rook may pass through an attacked square, because the rule concerns the king.

A rook that has moved previously and returned to its original square cannot be used for castling. Likewise, a king that has moved and returned cannot castle.

8. En Passant

En passant is a special pawn capture with a strict timing requirement.

- A pawn moves two squares forward from its starting position in one move.
- After that move, an opposing pawn is immediately adjacent to the moved pawn on the same rank and could have captured it had it moved only one square.
- The opposing pawn may capture en passant only on its immediately following move. The right disappears if any other move is made first.
- The capturing pawn moves diagonally forward to the square the two-square pawn passed over.
- The pawn that moved two squares is removed from the board, even though the capturing pawn does not land on its square.
- The en passant move is legal only if, after applying the capture, the moving player's own king is not in check.

9. Pawn Promotion

Promotion occurs when a pawn reaches the farthest rank from its starting side: White reaches the eighth rank; Black reaches the first rank.

- The pawn must immediately be replaced by one of: **Queen, Rook, Bishop, or Knight**.
- The promoted piece takes effect immediately as part of the move.
- A pawn may not remain on the promotion rank.
- Promotion may occur on a normal pawn move, a normal pawn capture, or an en passant sequence only insofar as the resulting pawn reaches the final rank (ordinary en passant itself does not promote because its capturing pawn advances one rank).

10. Captures

- **Normal captures:** A piece moves to a square occupied by an opposing piece according to its movement rules; the opposing piece is removed.
- **Special captures:** En passant removes the captured pawn from a different square than the capturing pawn's destination.
- A piece may not capture a friendly piece.
- The captured piece is removed before the resulting position is evaluated.
- The king may capture an opposing piece only when the destination square is not attacked by any opposing piece.

11. Draw Conditions

Stalemate

Automatic game-state condition: no legal moves for the side to move, and that side is not in check. Result: draw.

Insufficient mating material

A draw occurs when the position cannot possibly be checkmate under the applicable rules. Common examples include king versus king and king plus bishop versus king, or king plus knight versus king. Implementations should use a ruleset that correctly determines whether checkmate is still possible from the actual position, rather than relying only on a small material list.

Threefold repetition

A player may claim a draw when the same position has occurred at least three times. For repetition, the relevant position must be the same with respect to the side to move, the pieces on the same squares, castling rights, and en passant availability. Merely having the same piece placement is not sufficient if those state components differ. Threefold repetition is a claim-based draw under standard chess rules, so an implementation should distinguish the claim from an automatically declared game-over state unless its chosen ruleset explicitly automates the claim.

Fifty-move rule

A player may claim a draw when the last 50 moves by each side—100 half-moves/plies—have occurred without any pawn move or capture. This is a claim-based draw under standard chess rules. The half-move clock must reset after every pawn move or capture.

Draw by agreement

Players may agree to a draw. This is an optional UI action rather than an automatic position evaluation. A Draw button may offer the action, but it must not be confused with automatic detection of stalemate or other rule-based draw conditions.

Implementation note: If the application is intended to implement a particular tournament ruleset, define whether claimable draws are user-claimable, automatically offered, or automatically declared. The core position evaluator should keep objective state facts separate from UI actions.

12. Game State

A practical implementation can represent the game using a board position plus explicit status fields. The following states should be distinguishable:

State	Meaning
White to move	White is the active player and the game has not ended.
Black to move	Black is the active player and the game has not ended.
Check	The side to move's king is attacked. The game continues if at least one legal escape exists.
Checkmate	The side to move is in check and has no legal moves. Game over; the other side wins.
Stalemate	The side to move is not in check and has no legal moves. Game over; draw.
Draw	Game ended as a draw under an applicable draw condition or player agreement.
Game over	No further moves may be accepted. The terminal result must be preserved.

13. Implementation Checklist

- Legal movement
- Captures
- Check detection
- Checkmate detection
- Stalemate detection
- Castling
- En passant
- Promotion
- Draw detection
- Turn switching
- Game termination

Implementation note: This reference intentionally excludes AI behavior, online multiplayer, and website-specific design requirements. Its purpose is to define chess rules and game-state behavior for a local two-player implementation.